

Sigma HR System

API Documentation

Complete Reference for HR Module Endpoints

Employees | Attendance | Leave | Leave Types | Payroll

Version 1.0 — 2025

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1. Employee Management

Base URL: </api/HR/Employee> | Authentication: JWT Bearer Token required for most endpoints.

GET </api/HR/Employee>

Get All Employees

Returns a paginated list of all employees with optional search by name or employee number.

Parameters

Parameter	Type	Required	Description
searchName	string	No	Filter by Arabic name, English name, or employee number
page	integer	No	Page number (default: 1)
pageSize	integer	No	Number of records per page (default: 10)

Response

200 OK — *PagedResponse* containing list of *EmployeeDto* with fields: *Id*, *EmployeeNumber*, *NameAr*, *NameEn*, *Email*, *IdNumber*, *MobileNumber*, *JobId*, *JobName*, *DepartmentId*, *DepartmentName*, *NationalityId*, *NationalityName*, *HiringDate*, *IsActive*, *BankName*, *BankAccountNumber*, *IBAN*, *UserName*.

GET </api/HR/Employee/{id}>

Get Employee By ID

Returns full details of a single employee including salary breakdown.

Parameters

Parameter	Type	Required	Description
id	Guid	Yes	Employee unique identifier

Response

200 OK — *EmployeeCurrentDto* with all base fields plus salary details: *BasicSalary*, *HousingAllowance*, *MobilityAllowance*, *OtherAllowances*, *TotalSalary*. 404 Not Found if employee does not exist.

POST /api/HR/Employee**Create New Employee**

Creates a new employee record. Automatically creates an Identity user account, assigns the 'Employee' role, and initialises leave balances for all active leave types. Runs inside a database transaction.

Request Body (JSON)

Parameter	Type	Required	Description
EmployeeNumber	string	No	Unique employee number
NameAr	string	No	Employee name in Arabic
NameEn	string	No	Employee name in English
Email	string	Yes	Employee email — must be unique
IdNumber	string	No	National ID number
MobileNumber	string	No	Mobile phone number
JobId	Guid?	No	Reference to job/position
DepartmentId	Guid?	No	Reference to department
NationalityId	Guid?	No	Reference to nationality
HiringDate	DateTime	No	Date of hiring
BasicSalary	decimal?	No	Basic monthly salary
HousingAllowance	decimal?	No	Housing allowance
MobilityAllowance	decimal?	No	Mobility/transport allowance
OtherAllowances	decimal?	No	Other allowances
IsActive	bool	No	Employee active status
BankName	string	No	Bank name
BankAccountNumber	string	No	Bank account number
IBAN	string	No	IBAN number
UserName	string	No	Login username

Response

200 OK — 'Employee created successfully'. 400 if email already exists. 404 if JobId or DepartmentId invalid.

Note: A random 10-character password is auto-generated for the new user account.

PUT `/api/HR/Employee/{id}`

Update Employee

Updates employee data. Email and UserId cannot be changed after creation.

Parameters

Parameter	Type	Required	Description
id	Guid	Yes	Employee unique identifier

Request Body (JSON)

Parameter	Type	Required	Description
EmployeeNumber	string	No	Unique employee number
NameAr	string	No	Employee name in Arabic
NameEn	string	No	Employee name in English
IdNumber	string	No	National ID number
MobileNumber	string	No	Mobile phone number
JobId	Guid?	No	Reference to job/position
DepartmentId	Guid?	No	Reference to department
NationalityId	Guid?	No	Reference to nationality
HiringDate	DateTime	No	Date of hiring
BasicSalary	decimal?	No	Basic monthly salary
HousingAllowance	decimal?	No	Housing allowance
MobilityAllowance	decimal?	No	Mobility allowance
OtherAllowances	decimal?	No	Other allowances
IsActive	bool	No	Active status

Response

200 OK — 'Employee updated successfully'. 404 if not found.



/api/HR/Employee/{id}

Deactivate Employee (Soft Delete)

Sets IsActive = false. The record is NOT physically deleted from the database.

Parameters

Parameter	Type	Required	Description
id	Guid	Yes	Employee unique identifier

Response

200 OK — 'Employee deleted successfully'. 404 if not found.

Note: This is a soft delete — the employee record remains in the database.



/api/HR/Employee/{id}/reset-password

Reset Employee Password

Resets the employee's account password to a system default. Planned: send new password via WhatsApp notification.

Parameters

Parameter	Type	Required	Description
id	Guid	Yes	Employee unique identifier

Response

200 OK — 'Employee password reset successfully'. 404 if not found. 500 if reset fails.

Note: Currently resets to 'DefaultPassword123!'. WhatsApp notification integration is planned.

2. Attendance Management

Base URL: </api/HR/Attendance> | Authentication: JWT Bearer Token required for all endpoints.

POST </api/HR/Attendance/CheckIn>

Employee Check-In

Records the check-in time for the currently authenticated employee. The employee is identified automatically from the JWT token.

Response

200 OK — Check-in recorded. Returns error if already checked in today.

Note: No request body required. Employee ID is resolved from the authenticated user's token.

POST </api/HR/Attendance/CheckOut>

Employee Check-Out

Records the check-out time and calculates late minutes and overtime minutes based on shift schedule.

Response

200 OK — Check-out recorded with calculated durations. Returns error if no check-in found.

Note: System automatically calculates LateMinutes and OvertimeMinutes used in payroll generation.

POST </api/HR/Attendance/Filter>

Filter Attendance Records

Returns attendance records filtered by various criteria such as employee, date range, and status.

Request Body (JSON)

Parameter	Type	Required	Description
EmployeeId	Guid?	No	Filter by specific employee
FromDate	DateTime	No	Start of date range
ToDate	DateTime	No	End of date range
Status	string	No	Attendance status: Present, Absent, Late, etc.

Response

200 OK — List of attendance records matching the filter criteria.

3. Leave Management

Base URL: </api/HR/Leave> | Authentication: JWT Bearer Token required. Status flow: Pending → Approved / Rejected / Cancelled.

POST </api/HR/Leave>

Create Leave Request

Submits a new leave request for the authenticated employee. Validates date range, balance availability, and overlap with existing requests.

Request Body (JSON)

Parameter	Type	Required	Description
LeaveTypeId	Guid	Yes	ID of the leave type
FromDate	DateTime	Yes	Start date of the leave
ToDate	DateTime	Yes	End date of the leave
reason	string	No	Reason for the leave

Response

200 OK — 'Leave request created successfully'. Errors: employee/leave-type not found, dates in past, date overlap, insufficient balance.

Note: Dates cannot be in the past. FromDate must be <= ToDate. System checks for overlapping approved/pending requests.

PUT </api/HR/Leave/{requestId}/approve>

Approve Leave Request

Approves a pending leave request. Deducts the leave days from the employee's balance. Only pending requests can be approved.

Parameters

Parameter	Type	Required	Description
requestId	Guid	Yes	Leave request unique identifier

Response

200 OK — 'Leave request approved successfully'. Errors: request not found, not pending, insufficient balance.

Note: Updates LeaveBalance.UsedBalance and sets ApprovedById and ApprovedAt on the request.

PUT **/api/HR/Leave/{requestId}/reject**

Reject Leave Request

Rejects a pending leave request. Balance is not affected. Only pending requests can be rejected.

Parameters

Parameter	Type	Required	Description
requestId	Guid	Yes	Leave request unique identifier

Response

200 OK — 'Leave request rejected successfully'. 404 if not found. Error if not in Pending status.

PUT **/api/HR/Leave/{requestId}/cancel**

Cancel Leave Request

Cancels a leave request. If the request was already approved, the used balance days are restored to the employee's balance.

Parameters

Parameter	Type	Required	Description
requestId	Guid	Yes	Leave request unique identifier

Response

200 OK — 'Leave request cancelled successfully'. Error if already cancelled.

Note: For approved requests: UsedBalance is decremented (minimum 0). For pending requests: cancelled directly with no balance impact.

GET **/api/HR/Leave**

Get All Leave Requests

Returns all leave requests in the system (admin view). Includes status, dates, days count, and approval information.

Response

200 OK — List of LeaveRequestDto: Id, EmployeeId, LeaveTypeId, FromDate, ToDate, DaysCount, reason, Status, ApprovedAt, ApprovalComment.

GET **/api/HR/Leave/balance/{leaveTypeId}**

Get Leave Balance

Returns the remaining leave balance for the authenticated employee for a specific leave type in the current year.

Parameters

Parameter	Type	Required	Description
leaveTypeId	Guid	Yes	Leave type unique identifier

Response

200 OK — JSON object: { Balance: decimal }. Returns 0 if no balance record found.

Note: Balance = TotalBalance - UsedBalance for the current calendar year.

4. Leave Types Management

Base URL: </api/HR/LeaveType> | Defines the types of leave available (e.g. Annual, Sick, Emergency). No authentication attribute in controller — verify with team.

POST </api/HR/LeaveType>

Create Leave Type

Creates a new leave type. Name must be unique across all leave types.

Request Body (JSON)

Parameter	Type	Required	Description
Name	string	Yes	Unique name for the leave type (e.g. 'Annual Leave')
DefaultDaysPerYear	decimal	Yes	Default number of days allocated per year
IsActive	bool	Yes	Whether this leave type is currently active
AllowCarryForward	bool	Yes	Whether unused days can be carried to next year
IsPaid	bool	Yes	Whether this leave type is paid or unpaid

Response

201 Created — 'Leave type created successfully'. 400 if name already exists.

Note: When a new Employee is created, balances are auto-initialised for all active leave types.

GET </api/HR/LeaveType>

Get All Leave Types

Returns all leave types with their configuration.

Response

200 OK — List of LeaveTypeDto: Id, Name, DefaultDaysPerYear, IsActive, AllowCarryForward, IsPaid.

GET </api/HR/LeaveType/{id}>

Get Leave Type By ID

Returns a single leave type by its ID.

Parameters

Parameter	Type	Required	Description
id	Guid	Yes	Leave type unique identifier

Response

200 OK — LeaveTypeDto. 404 Not Found if leave type does not exist.

PUT `/api/HR/LeaveType/{id}`

Update Leave Type

Updates an existing leave type configuration. All fields are replaced.

Parameters

Parameter	Type	Required	Description
id	Guid	Yes	Leave type unique identifier

Request Body (JSON)

Parameter	Type	Required	Description
Name	string	Yes	Leave type name
DefaultDaysPerYear	decimal	Yes	Default days per year
IsActive	bool	Yes	Active status
AllowCarryForward	bool	Yes	Allow carry forward
IsPaid	bool	Yes	Is paid leave

Response

200 OK — 'Leave type updated successfully'. 404 if not found.

Note: Duplicate name check is currently commented out in the service — all names accepted on update.

DELETE `/api/HR/LeaveType/{id}`

Delete Leave Type

Permanently deletes a leave type. This is a hard delete — use with caution in production.

Parameters

Parameter	Type	Required	Description
id	Guid	Yes	Leave type unique identifier

Response

200 OK — 'Leave type deleted successfully'. 404 if not found.

Note: Hard delete — record is permanently removed. Consider deactivating (IsActive=false) instead.

5. Payroll Management

Base URL: `/api/HR/Payroll` | Payroll is calculated per month/year. Formula: $\text{NetSalary} = \text{BaseSalary} + \text{Overtime} - \text{LateDeduction} - \text{AbsenceDeduction} - \text{LeaveDeduction} + \text{Bonus} - \text{AdditionalDeduction}$.

POST `/api/HR/Payroll/generate`

Generate Monthly Payroll

Generates payroll for all active employees for the specified month and year. Calculates salary, deductions, and bonuses based on attendance and approved leaves.

Request Body (JSON)

Parameter	Type	Required	Description
Month	integer	Yes	Month number (1-12)
Year	integer	Yes	4-digit year (e.g. 2025)

Response

200 OK — 'Payroll generated successfully'. 400 if payroll already exists for that period.

Note: Calculation breakdown per employee: - $\text{LateDeduction} = (\text{TotalLateMinutes} / 60) * \text{LatePenaltyPerHour}$ - $\text{OvertimeAmount} = (\text{TotalOvertimeMinutes} / 60) * \text{OvertimeHourRate}$ - $\text{AbsenceDeduction} = \text{AbsentDays} * (\text{BasicSalary} / 30)$ - $\text{LeaveDeduction} = \text{UnpaidLeaveDays} * (\text{BasicSalary} / 30)$

GET `/api/HR/Payroll`

Get Payroll for Month/Year

Returns the full payroll details for a specific month and year, including per-employee breakdown.

Parameters

Parameter	Type	Required	Description
month	integer	Yes	Month number (1-12)
year	integer	Yes	4-digit year

Response

200 OK — List of `PayrollEmployeeDto`: `EmployeeId`, `EmployeeName`, `BaseSalary`, `OvertimeAmount`, `LateDeduction`, `AbsenceDeduction`, `LeaveDeduction`, `Bonus`, `AdditionalDeduction`, `NetSalary`. 404 if payroll not found.

PUT `/api/HR/Payroll/close/{id}`

Close Payroll Run

Marks a payroll run as closed (IsClosed = true). A closed payroll cannot be reopened.

Parameters

Parameter	Type	Required	Description
id	Guid	Yes	PayrollRun unique identifier

Response

200 OK — 'Payroll run closed successfully'. 404 if not found. 400 if already closed.

Note: Once closed, the payroll run is locked. Any corrections require creating a new run.

GET `/api/HR/Payroll/export`

Export Payroll to Excel

Generates and downloads an Excel file (.xlsx) with the payroll data for the specified month and year.

Parameters

Parameter	Type	Required	Description
month	integer	Yes	Month number (1-12)
year	integer	Yes	4-digit year

Response

200 OK — Binary Excel file download: Payroll-{month}-{year}.xlsx Columns: Employee, Base Salary, Overtime, Late Deduction, Absence Deduction, Leave Deduction, Bonus, Additional Deduction, Net Salary.

Note: Returns application/vnd.openxmlformats-officedocument.spreadsheetml.sheet content type.

Appendix: End-to-End Workflow Summary

New Employee Onboarding

1	POST /api/HR/Employee — Create employee (auto-creates user account + leave balances)
2	GET /api/HR/Employee/{id} — Verify employee data
3	PUT /api/HR/Employee/{id}/reset-password — Send password to employee

Daily Attendance

1	POST /api/HR/Attendance/CheckIn — Employee checks in at start of shift
2	POST /api/HR/Attendance/CheckOut — Employee checks out (late/overtime auto-calculated)
3	POST /api/HR/Attendance/Filter — HR views attendance records by date/employee

Leave Request Lifecycle

1	GET /api/HR/Leave/balance/{leaveTypeId} — Employee checks available balance
2	POST /api/HR/Leave — Employee submits leave request
3	GET /api/HR/Leave — HR reviews all pending requests
4	PUT /api/HR/Leave/{id}/approve OR /reject — HR takes action
5	PUT /api/HR/Leave/{id}/cancel — Employee or HR cancels if needed

Monthly Payroll Processing

1	POST /api/HR/Payroll/generate — Generate payroll (uses attendance + leave data)
2	GET /api/HR/Payroll?month=X&year;=Y — Review payroll details
3	GET /api/HR/Payroll/export?month=X&year;=Y — Export Excel report
4	PUT /api/HR/Payroll/close/{id} — Lock the payroll run

All endpoints return `ApiResponse<T>` with fields: `StatusCode`, `Data`, `Message`. Authenticated endpoints require Authorization: Bearer <token> header.